

Alok Kumar Mishra

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EDUCATION

- Indian Institute of Information Technology, Jabalpur
Bachelor of Technology - Electronics & Communication Engineering
Madhya Pradesh, India
Aug 2024 – Present

EXPERIENCE

- DigiLocker (MeitY), Government of India
Security Contributor (AppSec)
Remote
Feb 2025 – Apr 2025
 - Application Security & Android RE: Responsibly disclosed 10+ P1/P2 vulnerabilities in DigiLocker’s Android app (500M+ users) via dynamic/static analysis using Frida, JEB, Radare2.
 - Remediation & Recognition: Worked with DigiLocker engineers on fixes through detailed reports; received official Govt. of India acknowledgment for responsible disclosure.
- Rizin Project (Open-Source RE Framework)
Core Contributor – RE Tooling
Remote
Nov 2025 – Dec 2025
 - Binary Hardening: Improved ELF parsing robustness against malformed binaries and DoS vectors using bounds checks aligned with CVE-style hardening.
 - Core Tooling: Implemented De Bruijn decoding shortcuts and unified address APIs across core modules; added regression test binaries and RE documentation.

PROJECTS

- RzWeb – Rizin on Web (WebAssembly Reverse Engineering Platform)
WebAssembly (WASM), WASI, TypeScript, Python, Shell, Emscripten
GitHub
 - Built RzWeb, a browser-native reverse engineering IDE by compiling the Rizin toolchain to WebAssembly (WASM) (WASI/Emscripten), enabling secure static analysis and inspection directly in the web.
 - Engineered RzWasi execution and sandbox-safe host bindings to run Rizin modules inside a WASI runtime, solving syscall/runtime constraints and enabling reproducible builds across environments.
 - Implemented core RE workflows (disassembly, interactive console, graph navigation) with a responsive TypeScript UI, optimized for instant loading and near-instant analysis on cached smaller binaries.
 - Shipped a production-grade live web deployment; achieved 500+ stars, 50+ forks, and 100K+ visits in the first week itself, reducing setup friction from hours to under 2 minutes.
- Arthyx – Quant Finance Document Intelligence
Next.js, TypeScript, Neo4j, Vector Search
GitHub
 - Engineered a RAG-based quant system for deep analysis of 500+ page financial PDFs (Basel III, NPA, CRAR).
 - Modeled financial entities and risk propagation using Neo4j knowledge graphs, reducing manual review by 40%.
- HexaCiphers – Social Media Intelligence & OSINT
React, Python, NLP
GitHub
 - Built a real-time multilingual OSINT platform for Twitter/X campaign and narrative analysis.
 - Implemented bot detection and sentiment pipelines achieving 80% accuracy for cyber threat intelligence use cases.

TECHNICAL SKILLS

- Languages: Python, C/C++, Java, JavaScript, TypeScript, Bash, SQL, Smali, HTML/CSS
- Cybersecurity / RE: VAPT, Bug Bounty, AppSec, Android RE, Frida, Ghidra, IDA Pro, JEB, Radare2, Burp, Metasploit
- Web / Backend: React, Next.js, Node.js, FastAPI, Firebase, Tailwind, REST APIs
- AI / Finance: RAG Systems, NLP, Vector Search, Neo4j, Gemini API, Basel III, NPA/CRAR
- Tools: Git, Docker, Linux, Android Studio, Postman, Vercel

ACHIEVEMENTS

- Acknowledged Security Contributor by DigiLocker (GOI) for 10+ critical vulnerabilities (500M+ users).
- 1st Place – Can You Hack It Hackathon (200+ students), Team Lead.
- Top 20 / 700+ teams – Hack To Future 2025, IIT Gandhinagar.
- JEE Main Top 1.2%; CBSE Merit (100/100 CS); Global Rank 385 in CodeChef Starters 176C.

VOLUNTEER EXPERIENCE

- Jagrati – Non-Profit Education Initiative
Mathematics Tutor (Volunteer)
Jabalpur, India
Aug 2024 – Present
 - STEM Education Outreach: Providing free mathematics tutoring to underprivileged students preparing for Navodaya entrance examinations; designed simplified study plans to improve conceptual understanding.
- TechFest “AbhiKalpan” – IIITDM Jabalpur
Student Volunteer
Jabalpur, India
Apr 2025
 - Event Operations & Coordination: Assisted in planning and execution of the institute’s annual technical festival; managed logistics and on-site coordination for the Laser Constellation event, ensuring smooth participant flow.